

Slide 1: CardioTwin AI

A living digital representation of the patient for earlier, personalized cardiovascular risk awareness.

- Challenge: Digital Twin Challenge 2026 by Happiest Health
- Core Objective: Moving healthcare from episodic reaction to continuous proactive trajectory awareness.
- Key Innovation: N=1 personal physiological baseline modeling combined with multimodal EHR-wearable fusion.

Slide 2: The Clinical Problem

Why reactive cardiovascular care fails high-risk patients.

- Cardiovascular diseases represent the leading cause of global morbidity and mortality.
- High-risk patients (hypertension, T2D, dyslipidemia) are monitored episodically every 3-6 months.
- In between clinic visits, acute autonomic and hemodynamic decompensation develops silently.
- Traditional AI models evaluate static EHR snapshots once every few months — blind to acute drift.

Slide 3: Our Solution — The Living Digital Twin

Not just a risk score, but a living virtual physiological counterpart.

- Personalized Baseline Engine: Learns each patient's individual normal range (mean $\pm$ 2 SD).
- Twin Drift Score (0-100): Quantifies multivariate distance from personal homeostatic equilibrium.
- Multi-Horizon Forecasting: Dynamically updates 6h, 24h, and 72h deterioration risks.
- Bi-directional What-If Simulation: Clinicians test interventions before prescribing.

Slide 4: System Architecture & Ingestion

High-throughput, event-driven multimodal pipeline.

- Data Quality Engine: Validates optical PPG perfusion, motion artifacts, and flatlines.
- Multimodal Fusion: Combines static chronic vulnerability (EHR) with dynamic acute strain (wearables).
- Time-Series Integrity: Backward-looking rolling windows strictly isolated from future target labels.
- Zero Data Leakage: Complete patient-stratified cohort isolation across train/test splits.

Slide 5: Patient A-1042 Journey

The deterministic 24-hour deterioration progression.

- 08:00 Normal baseline: RHR 68 bpm, HRV 52 ms, Sleep 7.1h, Steps 6200.
- 10:00 to 12:00 Early deviation: RHR climbs to 75 bpm, HRV drops to 43 ms (vagal withdrawal).
- 14:00 Compounding deficit: 1.7h sleep deficit detected, ambulatory step count drops 45%.
- 16:00 to 18:00 Risk threshold crossed: RHR 82 bpm (+20.6%), Twin Drift hits 81 (High). Automated alert dispatched.

Slide 6: What-If Digital Twin Simulator

Counterfactual scenario modeling for clinician decision support.

- Interactive Clinician Knobs: Sleep duration, medication adherence, physical mobilization, stress reduction.
- Trajectory Divergence: Unmitigated deterioration trajectory (78% risk) vs. Simulated intervention (28% risk).
- Counterfactual Explainability: 'Largest modeled risk reduction (+30%) stems from restoring sleep toward baseline.'
- Mandatory Safety Invariant: Watermarked 'SIMULATED SCENARIO — NOT A CLINICAL PREDICTION'.

Slide 7: Model Evaluation & Ablation

Rigorous validation proving multimodal fusion superiority.

- Model A (EHR-Only): AUROC 0.487, AUPRC 0.209, Lead Time 2.1 hours (blind to acute decompensation).
- Model B (Wearable-Only): AUROC 0.648, AUPRC 0.380, Lead Time 7.2 hours.
- Model C (Digital Twin Fusion): AUROC 0.715, AUPRC 0.482, Lead Time 14.5 hours!
- Clinical Utility: False alerts reduced to 1.12 per patient-week, minimizing clinician alarm fatigue.

Slide 8: Robustness & Stress Testing

Resilience under missing wearable telemetry.

- 10% Missing Telemetry: AUROC drops by only 2.1%.
- 25% Missing Telemetry: AUROC retained at 0.694 (-6.7% degradation).
- 50% Severe Sensor Dropout: System defaults gracefully to personal baseline priors.
- Engineering Maturity: Models do not crash or collapse when sensors detach.

Slide 9: India-Scale Deployment Architecture

[Ayushman Bharat Digital Mission \(ABDM\) architectural readiness.](#)

- HL7 FHIR R4: Resources mapped to Patient, Observation (LOINC 8867-4), and Condition.
- ABHA Health ID Binding: Simulated ABHA address linking for seamless longitudinal continuity.
- Edge Baseline Caching: Personalized parameters computed and cached for scalable sub-second latency.
- Notice: ABDM/FHIR alignment is an architectural direction, not official government certification.

Slide 10: Safety, Ethics & Future Vision

Responsible AI principles and prospective clinical roadmap.

- Human-in-the-Loop: Never replaces clinical judgment; alerts are calm, explainable recommendations.
- Privacy & Security: 100% synthetic cohorts; production architecture includes RBAC and audit logging.
- Future Roadmap: Multi-center prospective observational registry with clinical PPG smartwatches.
- Vision: Proactive cardiovascular preservation for millions of high-risk individuals.